

AAE 590: Lie Group Methods for Control and Estimation

$SE_2(3)$: Extended Pose for Inertial Navigation

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Learning Objectives:

- Understand $SE_2(3)$ as the extended pose group
- Master the matrix representation and group operations
- Derive exponential/logarithm, adjoint, and Jacobians
- Apply to IMU-driven inertial navigation

Why Extend $SE(3)$?

The Problem

IMU propagation involves rotation, position, **and velocity**. $SE(3)$ only captures pose—we need a bigger group!

Inertial Navigation State

Standard INS state:

- $R \in SO(3)$: orientation (body to world)
- $\mathbf{v} \in \mathbb{R}^3$: velocity (world frame)
- $\mathbf{p} \in \mathbb{R}^3$: position (world frame)

Total: 9 degrees of freedom

IMU measurements:

- $\tilde{\omega}$: angular velocity (body frame)
- $\tilde{\mathbf{a}}$: specific force/acceleration (body frame)

Question

Can we combine $(R, \mathbf{v}, \mathbf{p})$ into a single matrix Lie group?

Why Use a Lie Group for INS?

Traditional EKF approach:

- State: $\mathbf{x} = (R, \mathbf{v}, \mathbf{p}, \mathbf{b}_g, \mathbf{b}_a)$
- Covariance: $P \in \mathbb{R}^{15 \times 15}$ (or more)
- Problem: Covariance is state-dependent!

Lie group approach (Invariant EKF):

- State: $X \in \text{SE}_2(3)$ (or extended group with biases)
- Error: Lives in Lie algebra
- Covariance: State-independent (for certain systems)!

Key Insight [barrau2017invariant]

When the dynamics have “group-affine” structure, the Invariant EKF has autonomous error dynamics—the linearization doesn’t depend on the state!

The Extended Special Euclidean Group $SE_2(3)$ [barrau2017invariant]

Definition

$SE_2(3) = SO(3) \ltimes (\mathbb{R}^3 \times \mathbb{R}^3)$ is the group of **double direct spatial isometries**.

Matrix representation (5×5):

$$X = \begin{bmatrix} R & \mathbf{v} & \mathbf{p} \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix} \in \mathbb{R}^{5 \times 5}$$

where:

- $R \in SO(3)$: rotation matrix
- $\mathbf{v} \in \mathbb{R}^3$: velocity (world frame)
- $\mathbf{p} \in \mathbb{R}^3$: position (world frame)

Dimension: 9 (3 rotation + 3 velocity + 3 position)

SE₂(3) Group Operations

Composition:

$$X_1 X_2 = \begin{bmatrix} R_1 R_2 & R_1 \mathbf{v}_2 + \mathbf{v}_1 & R_1 \mathbf{p}_2 + \mathbf{p}_1 \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix}$$

Inverse:

$$X^{-1} = \begin{bmatrix} R^T & -R^T \mathbf{v} & -R^T \mathbf{p} \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix}$$

Identity:

$$I = \begin{bmatrix} I_{3 \times 3} & \mathbf{0} & \mathbf{0} \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix}$$

Semi-Direct Product Structure

$$\text{SE}_2(3) = \text{SO}(3) \ltimes (\mathbb{R}^3 \times \mathbb{R}^3)$$

The multiplication rule shows:

$$(R_1, \mathbf{v}_1, \mathbf{p}_1) \cdot (R_2, \mathbf{v}_2, \mathbf{p}_2) = (R_1 R_2, R_1 \mathbf{v}_2 + \mathbf{v}_1, R_1 \mathbf{p}_2 + \mathbf{p}_1)$$

Key insight: R acts on **both** $\mathbf{v}$ and $\mathbf{p}$ independently!

Compare to SE(3):

- $\text{SE}(3) = \text{SO}(3) \ltimes \mathbb{R}^3$
- Position coupled to rotation
- 6 DOF

SE₂(3):

- $\text{SE}_2(3) = \text{SO}(3) \ltimes (\mathbb{R}^3 \times \mathbb{R}^3)$
- Velocity and position independent
- 9 DOF

$SE_2(3)$ vs. $SE(3)$: Embedding

$SE(3)$ **embeds naturally into** $SE_2(3)$:

For $X = \begin{bmatrix} R & \mathbf{p} \\ \mathbf{0}^T & 1 \end{bmatrix} \in SE(3)$, define:

$$\iota(X) = \begin{bmatrix} R & \mathbf{0} & \mathbf{p} \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix} \in SE_2(3)$$

This is a group homomorphism: $\iota(X_1 X_2) = \iota(X_1) \iota(X_2)$

Subgroups of $SE_2(3)$

- $SO(3) \subset SE_2(3)$ (rotations only)
- $SE(3) \subset SE_2(3)$ (pose without velocity)
- $\mathbb{R}^3 \times \mathbb{R}^3 \subset SE_2(3)$ (translations in $\mathbf{v}$ and $\mathbf{p}$)

The Lie Algebra $\mathfrak{se}_2(3)$

Definition

The Lie algebra $\mathfrak{se}_2(3) = T_I SE_2(3)$ consists of matrices:

$$\xi^\wedge = \begin{bmatrix} \omega^\wedge & \mathbf{a} & \mathbf{b} \\ \mathbf{0}^T & 0 & 0 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix} \in \mathbb{R}^{5 \times 5}$$

where:

- $\mathbf{a} \in \mathbb{R}^3$: “velocity” component (will relate to acceleration)
- $\mathbf{b} \in \mathbb{R}^3$: “position” component (will relate to velocity)
- $\omega \in \mathbb{R}^3$: angular velocity

Dimension: 9 (matches the group!)

Lie algebra vector:

$$\xi = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \\ \omega \end{bmatrix} \in \mathbb{R}^9$$

Wedge and Vee for $SE_2(3)$

Wedge: $\mathbb{R}^9 \rightarrow \mathfrak{se}_2(3)$

$$\xi^\wedge = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \\ \omega \end{bmatrix}^\wedge = \begin{bmatrix} \omega^\wedge & \mathbf{a} & \mathbf{b} \\ \mathbf{0}^T & 0 & 0 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix}$$

Vee: $\mathfrak{se}_2(3) \rightarrow \mathbb{R}^9$

$$\begin{bmatrix} \omega^\wedge & \mathbf{a} & \mathbf{b} \\ \mathbf{0}^T & 0 & 0 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix}^\vee = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \\ \omega \end{bmatrix}$$

Convention: $\xi = (\mathbf{a}^T, \mathbf{b}^T, \omega^T)^T$ (translation first)

Alternative conventions exist!

Some papers use rotation-first ordering. Always check the source.

The algebra $\mathfrak{se}_2(3)$ decomposes as:

$$\mathfrak{se}_2(3) = \mathfrak{so}(3) \oplus \mathbb{R}^3 \oplus \mathbb{R}^3$$

As vector spaces:

- Angular part: $\omega^\wedge \in \mathfrak{so}(3)$ (3D)
- First translation: $\mathbf{a} \in \mathbb{R}^3$ (3D)
- Second translation: $\mathbf{b} \in \mathbb{R}^3$ (3D)

Key difference from $\mathfrak{se}(3)$:

- $\mathfrak{se}(3)$: one translation coupled to rotation
- $\mathfrak{se}_2(3)$: two translations, both coupled to rotation, but not to each other

Exponential Map for $SE_2(3)$

Given $\xi = (\mathbf{a}^T, \mathbf{b}^T, \omega^T)^T$:

$$\text{Exp}(\xi^\wedge) = \begin{bmatrix} R & \mathbf{v} & \mathbf{p} \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix}$$

where:

$$R = \text{Exp}_{SO(3)}(\omega^\wedge) \quad (\text{Rodrigues' formula})$$

$$\mathbf{v} = J_l(\omega) \mathbf{a}$$

$$\mathbf{p} = J_l(\omega) \mathbf{b}$$

$J_l(\omega)$ is the left Jacobian of $SO(3)$:

$$J_l(\omega) = I + \frac{1 - \cos \theta}{\theta^2} \omega^\wedge + \frac{\theta - \sin \theta}{\theta^3} (\omega^\wedge)^2$$

Exponential: Derivation Sketch

Exponential series:

$$\text{Exp}(\xi^\wedge) = I + \xi^\wedge + \frac{(\xi^\wedge)^2}{2!} + \frac{(\xi^\wedge)^3}{3!} + \dots$$

Powers of $\xi^\wedge$:

$$(\xi^\wedge)^k = \begin{bmatrix} (\omega^\wedge)^k & (\omega^\wedge)^{k-1} \mathbf{a} & (\omega^\wedge)^{k-1} \mathbf{b} \\ \mathbf{0}^T & 0 & 0 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix}$$

Summing gives:

$$R = \sum_{k=0}^{\infty} \frac{(\omega^\wedge)^k}{k!} = \text{Exp}_{\text{SO}(3)}(\omega^\wedge)$$

$$\mathbf{v} = \sum_{k=0}^{\infty} \frac{(\omega^\wedge)^k}{(k+1)!} \mathbf{a} = J_I(\omega) \mathbf{a}$$

Same structure for $\mathbf{p}$ with $\mathbf{b}$.

Exponential: Special Cases

Pure translation ($\omega = 0$):

$$\text{Exp} \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \\ \mathbf{0} \end{pmatrix}^{\wedge} = \begin{bmatrix} I & \mathbf{a} & \mathbf{b} \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix}$$

(Note: $J_I(\mathbf{0}) = I$, so $\mathbf{v} = \mathbf{a}$ and $\mathbf{p} = \mathbf{b}$)

Pure rotation ($\mathbf{a} = \mathbf{b} = \mathbf{0}$):

$$\text{Exp} \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \\ \omega \end{pmatrix}^{\wedge} = \begin{bmatrix} \text{Exp}_{\text{SO}(3)}(\omega^{\wedge}) & \mathbf{0} & \mathbf{0} \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix}$$

Logarithmic Map for $SE_2(3)$

Given $X = \begin{bmatrix} R & \mathbf{v} & \mathbf{p} \\ \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & 0 & 1 \end{bmatrix}$:

Step 1: Extract rotation

$$\boldsymbol{\omega} = \text{Log}_{SO(3)}(R)^\vee$$

Step 2: Extract translations

$$\mathbf{a} = J_l^{-1}(\boldsymbol{\omega})\mathbf{v}$$

$$\mathbf{b} = J_l^{-1}(\boldsymbol{\omega})\mathbf{p}$$

Result: $\xi = (\mathbf{a}^T, \mathbf{b}^T, \boldsymbol{\omega}^T)^T$

Numerical Stability

Use Taylor series for J_l^{-1} when $\|\boldsymbol{\omega}\| < \epsilon$.

The Adjoint Representation for $\text{SE}_2(3)$

Definition

The adjoint $\text{Ad}_X : \mathfrak{se}_2(3) \rightarrow \mathfrak{se}_2(3)$:

$$\text{Ad}_X(\xi^\wedge) = X\xi^\wedge X^{-1}$$

Matrix form on $\mathbb{R}^9$:

$$\text{Ad}_X = \begin{bmatrix} R & \mathbf{0} & \mathbf{v}^\wedge R \\ \mathbf{0} & R & \mathbf{p}^\wedge R \\ \mathbf{0} & \mathbf{0} & R \end{bmatrix} \in \mathbb{R}^{9 \times 9}$$

Block structure:

- 3×3 blocks
- Block upper triangular (translation-first ordering)
- R appears on diagonal

Adjoint: Action on Lie Algebra

Action on twist:

$$\text{Ad}_X \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \\ \boldsymbol{\omega} \end{bmatrix} = \begin{bmatrix} R\mathbf{a} + \mathbf{v}^\wedge R\boldsymbol{\omega} \\ R\mathbf{b} + \mathbf{p}^\wedge R\boldsymbol{\omega} \\ R\boldsymbol{\omega} \end{bmatrix}$$

Interpretation:

- Velocity part: rotated by R , plus coupling from rotation
- Position part: rotated by R , plus coupling from rotation
- Angular part: rotated by R

Frame Transformation

Ad_X transforms a Lie algebra element from one frame to another, properly accounting for the semi-direct product structure.

The Lie Bracket for $SE_2(3)$

For $\xi_1 = (\mathbf{a}_1^T, \mathbf{b}_1^T, \omega_1^T)^T$ and $\xi_2 = (\mathbf{a}_2^T, \mathbf{b}_2^T, \omega_2^T)^T$:

$$[\xi_1, \xi_2] = \begin{bmatrix} \omega_1 \times \mathbf{a}_2 - \omega_2 \times \mathbf{a}_1 \\ \omega_1 \times \mathbf{b}_2 - \omega_2 \times \mathbf{b}_1 \\ \omega_1 \times \omega_2 \end{bmatrix}$$

The small adjoint ad_ξ :

$$\text{ad}_\xi = \begin{bmatrix} \omega^\wedge & \mathbf{0} & \mathbf{a}^\wedge \\ \mathbf{0} & \omega^\wedge & \mathbf{b}^\wedge \\ \mathbf{0} & \mathbf{0} & \omega^\wedge \end{bmatrix}$$

Note: $\mathbf{a}$ and $\mathbf{b}$ don't interact with each other—only through ω !

Bracket Properties

Key observations:

1. **Angular bracket:** Same as $SO(3)$

$$[\omega_1, \omega_2]_\omega = \omega_1 \times \omega_2$$

2. **Velocity/position brackets:** Coupled to rotation only

$$[\mathbf{a}_1, \mathbf{a}_2]_a = 0, \quad [\mathbf{b}_1, \mathbf{b}_2]_b = 0$$

3. **Cross-coupling:**

$$[\omega, \mathbf{a}]_a = \omega \times \mathbf{a}, \quad [\omega, \mathbf{b}]_b = \omega \times \mathbf{b}$$

Structure

The translation parts $\mathbb{R}^3 \times \mathbb{R}^3$ form an abelian normal subgroup. The bracket captures how rotation “twists” translations.

Left and Right Jacobians for $SE_2(3)$

General form:

$$J_l(\xi) = \sum_{k=0}^{\infty} \frac{1}{(k+1)!} \text{ad}_{\xi}^k$$

Block structure:

$$J_l(\xi) = \begin{bmatrix} J_l^{\text{SO}(3)}(\omega) & \mathbf{0} & Q_a \\ \mathbf{0} & J_l^{\text{SO}(3)}(\omega) & Q_b \\ \mathbf{0} & \mathbf{0} & J_l^{\text{SO}(3)}(\omega) \end{bmatrix}$$

where $J_l^{\text{SO}(3)}(\omega)$ is the $SO(3)$ left Jacobian and Q_a, Q_b are coupling terms.

Right Jacobian: $J_r(\xi) = J_l(-\xi)$

Jacobian: Closed-Form Expression

The coupling terms Q_a and Q_b :

For $\theta = \|\omega\|$:

$$\begin{aligned} Q_a = & \frac{1}{2} \mathbf{a}^\wedge + c_1(\omega^\wedge \mathbf{a}^\wedge + \mathbf{a}^\wedge \omega^\wedge + \omega^\wedge \mathbf{a}^\wedge \omega^\wedge) \\ & + c_2(\omega^\wedge \omega^\wedge \mathbf{a}^\wedge + \mathbf{a}^\wedge \omega^\wedge \omega^\wedge - 3\omega^\wedge \mathbf{a}^\wedge \omega^\wedge) \\ & + c_3(\omega^\wedge \mathbf{a}^\wedge \omega^\wedge \omega^\wedge + \omega^\wedge \omega^\wedge \mathbf{a}^\wedge \omega^\wedge) \end{aligned}$$

Similar expression for Q_b with $\mathbf{b}$ replacing $\mathbf{a}$.

Coefficients:

$$c_1 = \frac{\theta - \sin \theta}{\theta^3}, \quad c_2 = \frac{\theta^2 + 2 \cos \theta - 2}{2\theta^4}, \quad c_3 = \frac{2\theta - 3 \sin \theta + \theta \cos \theta}{2\theta^5}$$

Small Angle Approximation

For $\|\xi\| \ll 1$:

$$J_l(\xi) \approx I + \frac{1}{2}\text{ad}_\xi + \frac{1}{6}\text{ad}_\xi^2$$

At identity:

$$J_l(0) = J_r(0) = I_{9 \times 9}$$

First-order approximation:

$$J_l(\xi) \approx I + \frac{1}{2} \begin{bmatrix} \omega^\wedge & \mathbf{0} & \mathbf{a}^\wedge \\ \mathbf{0} & \omega^\wedge & \mathbf{b}^\wedge \\ \mathbf{0} & \mathbf{0} & \omega^\wedge \end{bmatrix}$$

Practical Use

For IMU integration with small Δt , first-order approximation often suffices.

Measurements:

$$\tilde{\omega} = \omega + \mathbf{b}_g + \mathbf{n}_g$$

$$\tilde{\mathbf{a}} = R^T(\dot{\mathbf{v}} - \mathbf{g}) + \mathbf{b}_a + \mathbf{n}_a$$

where:

- $\mathbf{b}_g, \mathbf{b}_a$: slowly varying biases
- $\mathbf{n}_g, \mathbf{n}_a$: white noise
- $\mathbf{g} = (0, 0, -g)^T$: gravity in world frame

Continuous-time dynamics:

$$\dot{R} = R\omega^\wedge$$

$$\dot{\mathbf{v}} = R\mathbf{a} + \mathbf{g}$$

$$\dot{\mathbf{p}} = \mathbf{v}$$

where $\mathbf{a} = \tilde{\mathbf{a}} - \mathbf{b}_a - \mathbf{n}_a$ is the true specific force.

Dynamics on $SE_2(3)$: Mixed Invariant Form

The IMU dynamics on $SE_2(3)$: $\dot{X} = \xi_R^\wedge X + X \xi_L^\wedge$

$$\xi_L^\wedge = \begin{bmatrix} \omega^\wedge & \mathbf{a} & \mathbf{0} \\ \mathbf{0}^T & 0 & 1 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix}, \quad \xi_R^\wedge = \begin{bmatrix} \mathbf{0} & \mathbf{g} & \mathbf{0} \\ \mathbf{0}^T & 0 & -1 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix}$$

Expand to verify:

$$X \xi_L^\wedge = \begin{bmatrix} R\omega^\wedge & R\mathbf{a} & \mathbf{v} \\ \mathbf{0}^T & 0 & +1 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix}, \quad \xi_R^\wedge X = \begin{bmatrix} \mathbf{0} & \mathbf{g} & \mathbf{0} \\ \mathbf{0}^T & 0 & -1 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix}$$

$$\text{Sum: } \dot{X} = \begin{bmatrix} R\omega^\wedge & R\mathbf{a} + \mathbf{g} & \mathbf{v} \\ \mathbf{0}^T & 0 & 0 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix} \quad \checkmark \quad (\pm 1 \text{ artifacts cancel in row 4})$$

Key: The $+1$ in $\xi_L^\wedge$ copies column $\mathbf{v}$ of X into $\dot{\mathbf{p}}$. The -1 in $\xi_R^\wedge$ cancels the bookkeeping artifact. **Neither** $\xi_L^\wedge$ nor $\xi_R^\wedge$ is in $\mathfrak{se}_2(3)$!

Group-Affine Structure

The IMU dynamics are “group-affine” (mixed invariant):

$$\dot{X} = f_u(X) = \xi_R^\wedge X + X \xi_L^\wedge$$

Using $(M-C)X + X(N+C)$: physics $N = \begin{bmatrix} \omega^\wedge & \mathbf{a} & \mathbf{0} \\ \mathbf{0}^T & 0 & 0 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix} \in \mathfrak{se}_2(3), \quad M = \begin{bmatrix} \mathbf{0} & \mathbf{g} & \mathbf{0} \\ \mathbf{0}^T & 0 & 0 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix}$

Coupling: $C = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0}^T & 0 & 1 \\ \mathbf{0}^T & 0 & 0 \end{bmatrix} \notin \mathfrak{se}_2(3) \Rightarrow \xi_L^\wedge = N + C, \quad \xi_R^\wedge = M - C$

Key Property [barrau2017invariant]

For group-affine systems, the error dynamics (for left or right invariant error) are **autonomous**—they don't depend on the state X !

Consequence: The EKF linearization is state-independent. No need to re-linearize!

Discrete-Time Integration

Given: X_k , IMU measurements $(\tilde{\omega}_k, \tilde{\mathbf{a}}_k)$, time step Δt

Simplified integration (zeroth-order hold on inputs):

$$X_{k+1} = X_k \text{Exp}((\xi_k \Delta t)^\wedge) \cdot \Gamma(\mathbf{g}, \Delta t)$$

where:

$$\xi_k = \begin{bmatrix} \tilde{\mathbf{a}}_k - \mathbf{b}_a \\ \mathbf{0} \\ \tilde{\omega}_k - \mathbf{b}_g \end{bmatrix}, \quad \Gamma(\mathbf{g}, \Delta t) = \begin{bmatrix} I & \mathbf{g}\Delta t & \frac{1}{2}\mathbf{g}\Delta t^2 \\ \mathbf{0}^T & 1 & \Delta t \\ \mathbf{0}^T & 0 & 1 \end{bmatrix}$$

Γ accounts for gravity and $\dot{\mathbf{p}} = \mathbf{v}$ coupling.

Integration: Alternative Form

Exact integration (for constant inputs):

Let $\omega = \tilde{\omega} - \mathbf{b}_g$ and $\mathbf{a} = \tilde{\mathbf{a}} - \mathbf{b}_a$. Then:

$$\mathbf{R}_{k+1} = \mathbf{R}_k \text{Exp}_{\text{SO}(3)}(\omega \Delta t)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{R}_k J_I(\omega \Delta t) \mathbf{a} \Delta t + \mathbf{g} \Delta t$$

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t + \mathbf{R}_k Q(\omega \Delta t, \mathbf{a} \Delta t) + \frac{1}{2} \mathbf{g} \Delta t^2$$

where Q is the position integration term (involves J_I and higher-order terms).

Higher-Order Integration

For better accuracy, use midpoint rule or higher-order methods. The group structure is preserved by construction.

Error Definition on $\text{SE}_2(3)$

Left-invariant error:

$$\eta^L = X^{-1} \hat{X}$$

Right-invariant error:

$$\eta^R = \hat{X} X^{-1}$$

At perfect estimate: $\eta = I$

Linearized error (in Lie algebra):

$$\xi = \text{Log}(\eta)^\vee \in \mathbb{R}^9$$

Choice of Error Convention

The choice between left and right invariant error depends on the measurement model. For IMU + landmarks, left-invariant is typically preferred.

Invariant EKF: Key Result

For group-affine dynamics:

$$\dot{X} = \xi_R^\wedge X + X \xi_L^\wedge$$

The left-invariant error dynamics are:

$$\dot{\eta}^L = \eta^L \xi_L^\wedge - \xi_L^\wedge \eta^L + \xi_R^\wedge \eta^L - \eta^L \xi_R^\wedge = [\eta^L, \xi_L^\wedge] + [\xi_R^\wedge, \eta^L]$$

Linearization (to first order in ξ):

$$\dot{\xi} = A\xi + w$$

where A depends on $\xi_L^\wedge$ and $\xi_R^\wedge$, but **NOT** on X !

State-Independent Linearization

This is the magic of InEKF: the system matrix A doesn't depend on the state estimate, making the filter more robust.

Extended State with Biases

Full state for practical INS:

Augment $SE_2(3)$ with bias states:

$$\text{State} = (X, \mathbf{b}_g, \mathbf{b}_a) \in SE_2(3) \times \mathbb{R}^3 \times \mathbb{R}^3$$

Total dimension: $9 + 3 + 3 = 15$

Bias dynamics:

$$\dot{\mathbf{b}}_g = \mathbf{n}_{bg}$$

$$\dot{\mathbf{b}}_a = \mathbf{n}_{ba}$$

Error state:

$$\text{Error} = (\xi, \delta \mathbf{b}_g, \delta \mathbf{b}_a) \in \mathbb{R}^{15}$$

SE₂(3) Summary

Concept	SE ₂ (3)
Group	5×5 matrices $(R, \mathbf{v}, \mathbf{p})$
Structure	$\text{SO}(3) \ltimes (\mathbb{R}^3 \times \mathbb{R}^3)$
Lie algebra	$\xi = (\mathbf{a}^T, \mathbf{b}^T, \boldsymbol{\omega}^T)^T \in \mathbb{R}^9$
Dimension	9
$\text{Exp}(\xi^\wedge)$	Rodrigues + J_I for both $\mathbf{v}$ and $\mathbf{p}$
Ad_X	9×9 block upper triangular matrix
IMU dynamics	Group-affine: $\dot{X} = \xi_R X + X \xi_L$
Key property	Autonomous error dynamics (InEKF)

Looking Ahead

Week 8: Group-Affine Systems

- Formal definition and properties
- More examples beyond IMU
- Autonomous error dynamics proof

Week 9: Invariant EKF

- Full derivation of InEKF
- Measurement models
- Implementation details

Applications:

- Visual-inertial odometry
- GPS/INS fusion
- Legged robot state estimation

